2022 10th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)
Amity University, Noida, India. Oct 13-14, 2022

Pneumonia Detection from Chest X-ray Images using Transfer Learning

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***Abstract*— Pneumonia is a disease that an individual can acquire at any stage of the life. As per a report from Cleveland clinic, America, pneumonia is responsible for about 18% of all contagious diseases. In the phases that follow, this condition may lead to death. Chest radiographs have been found to as constant value by field professionals in clinical practice in order to identify pneumonia. In this work, chest X-ray scans, which are accessible for the identification and diagnosis of pneumonia are utilized. For feature extraction from the images, a VGG16 transfer learning model is used. In this paper, various pre-existing models for pneumonia detection have been reviewed along with the identification of their performance results. A framework of VGG16 used for the study has been explained with a variety of applications of deep learning in disease diagnosis. The dataset for the research is collected from kaggle containing 5,856 images, which have been further divided into training and testing datasets. Further, the results have been presented in the form of accuracy, precision, F1-score and recall as 90.8%, 0.9102, and 0.935, 0.9615, respectively.**

Keywords*— *Deep Learning, Convolutional Neural Network, VGG16, Pneumonia, Transfer Learning.

I. Introduction

Throughout history, chronic diseases and viruses have claimed countless lives and resulted in massive crises which lasted for years [1][2]. These diseases can spread throughout societies over a long period of time. A pandemic, on the other hand, is described as the emergence of illnesses, injuries, or many other health complications in which numbers of cases appear to be increased with a very less recovery rate. The epidemic is different from a pandemic in such a way that it is more localized and less likely to cause widespread panic. Whereas, a pandemic can spread worldwide and results in health issues in pregnant ladies which causes defects in the infants. One among the pandemic includes; pneumonia, which is a severe illness that creates lots of health complications. Pneumonia causes about 1 million hospitalizations in the U.S. yearly, with an approximately 1.5 million mortality rate [3]. Chest radiographs are the most prominent and commonly available method for identifying pneumonia, as these radiographs have a wide range of uses in healthcare and observational studies. Pneumonia diagnosis in chest radiographs is a difficult and time-consuming process and requires the involvement of well-experienced radiologists for the diagnosis. It is a pulmonary disease which affects the lungs and is caused by fungus, bacteria, and infections which can spread through the air [4]. As the condition worsens, pneumonia spreads to the lungs, creating greater respiratory issues. Pneumonia is one of the main causes of morbidity and death in kids under the age of five. A radiograph is the most reliable technique for identifying pneumonia in which radiologists methodically assess and prioritize radiographs which is a time-consuming process and may lead to controversy among doctors when it comes to prediction and diagnosis. As a result of this issue, pneumonia diagnosing techniques which are safe to implement in the medical industries for pneumonia diagnosis and prediction with no errors are required to be developed. Statistical approaches, machine learning (ML), along with deep learning (DL) are convenient and efficient techniques in health status detection systems [5][6]. These systems can be implemented to overcome highly complex visual tasks in the medical imaging industry, such as respiratory disease segmentation and classification. Recent advances in DL have aided in achieving, and possibly even greatly exceeding, human performance in a variety of activities by using transfer learning techniques [7]. These techniques make it possible to transfer the information from one model to another model for achieving the goal. Treatment outcomes, such as active presence studies and medical treatments, can be determined using DL-based transfer learning models. These techniques have been identified as the positive outcome representing methods in lung disease identification. With the pneumonia outbreak, DL has become a more significant application in healthcare as it can early detect pneumonia by analyzing radiographs and CT- scans [6]. Convolutional neural networks (CNNs) is a DL technique which has been identified as a well-suited model for processing images, such as MRI data, CT scans and X-rays. According to researchers at Stanford University, CNN models are built with the premise that they process images and allow networks to run more effectively and manage a huge amount of data [5]. As a consequence, when it comes to determining key characteristics in image studies, CNN sometimes surpasses – the efficiency of human professionals. It can also predict whether a person suffering from the disease requires ICU admission and also assist in the identification of prospective pneumonia patients. There are various architectures of CNN including, ImageNet, VGG, InceptionV2 etc. Where VGG is capable of extracting features from the images in a significant way and reduces the computational time. Hence, in this paper, the VGG16 transfer learning model has been used for the identification of pneumonia in chest X-ray images.

This work is organized as follows: Section II contains the information about pre-existing models to detect the presence of pneumonia. Section III defines the methodology of the work along with the dataset utilized for performing the experiment. Section IV compares the results of the VGG16 transfer learning model with the state-of-the-art models. Finally, results and future work is introduced in section V.

II. State of the art

Singh et al. [4] have presented a Quaternion CNN technique for diagnosing the presence of pneumonia in chest X-ray radiographs. Furthermore, for validating this model, a comparison was done among the results of ResNet and their proposed model, which identified Quaternion CNN as a better model as compared to ResNet model. Barhoom et al. [8] have presented a study for identifying the best performing Deep Learning algorithm for the identification of pneumonia. A chest X-ray dataset has been used for training, testing, and validating the model. The Adam optimizer and ReLu function were utilized for setting up the complete experiment in which various models including VGG, and ResNet have been introduced along with their merits and demerits. As a result, it has been identified that, in CNN, the VGG16 model performs better than other DL models.

Haghanifar et al. [8] have presented a CNN-based CXNet model for diagnosing pneumonia in Chest X-ray images dataset. The dataset, namely pediatric CXR, NIH CXR-14 and Tuberculosis, have been collected from numerous different sources, such as radiopaedia, the US national library of medicine etc. The database implemented in the study has a total number of 9.600 images including normal and pneumonia patients. Ravi et al. [10] have utilized and implemented an ensemble of two models namely, transfer learning and meta-classifier for identifying the presence pneumonia in Chest X-ray images. The meta-classifier also includes two models, specifically designed for prediction and classification as SVM and logistic regression, respectively. The results of the experiment have been mentioned in the form of F1-score as 0.0321. The dataset used in this study consisted of a total number of 5,856 images out of which 1583 were for normal and 4273 images were for pneumonia.

Zhang et al. [11] have presented an AlexNet model for predicting pneumonia in Chest X-ray scans. Along with this, batch normalization is used for reducing the covariance and increasing the speed of training of the model. As a result, the three AlexNet models were compared, in which DC-Net-R has been identified as the best model as compared to other models used in the network in terms of accuracy. Ayan et al. [12] have used an ensemble CNN framework for diagnosing pneumonia in which, firstly the experiment was performed to select three transfer learning models for creating an ensemble model for prediction. The model was trained and tested on a chest X-ray dataset with a learning rate of 0.0001, and Adam optimizer, which resulted in the model's accuracy as 90.71%, and sensitivity as 0.9776.

Rahimzadeh et al. [13] have presented a Deep CNN model by ensembling two CNN models namely, ResNet and XceptionNet models for predicting pneumonia. Ieracitano et al. [14] have used a CNN model for the prediction of pneumonia using chest X-ray radiographs. Various features have been extracted from the fuzzy and X-ray images and the results have been compared with various DL models to validate the developed model. The results of the study were mentioned in the form of accuracy as 81%.

Kundu et al. [15] have developed an ensemble model by combining DenseNet121, ResNet18 and GoogleNet feature extractor. This model has been applied to two different datasets namely, RSNA and Kermany's dataset collected from different sources. Finally, the results were mentioned and compared with various models to validate the ensemble model. Ayan et al. [16] have used a Chest X-ray dataset for predicting pneumonia using VGG16 and Xception model. The results of the performance have been mentioned in the form of accuracy where the VGG16 has achieved higher accuracy than Xception model.

III. Material and Methods

This section of the study introduces the VGG16 transfer learning model for the prediction of the presence of pneumonia in chest radiographs. Furthermore, the complete methodology for developing the model along with the details of the dataset is also mentioned. Transfer learning is a technique which is used for transferring information from one DL system to another [7]. It is the process of reusing a model that has already been trained on a different problem. With transfer learning, a computer can use its experience of one function to better generalize the solution to another problem. There are various pre-trained models including VGG16, VGG19, InceptionV3 and ImageNet.

The VGG16 is a feature extraction technique designed to identify the presence of pneumonia. In the year of 2014, after an ILSVR(Imagenet) competition, VGG16 became popular as a CNN framework due to its merits over other transfer learning models. It has been found that CNN is one of the best classification models. This model is remarkable because it has 3x3 filter convolution and instead of setting a large number of hyper-parameters, it employs the exact padding and max pooling layer of a 2x2 filter. Throughout the complete architecture, the pooling and convolutional layers are arranged identically. Eventually, the two fully connected and softmax layers are used at the end for predicting the output [17][18]. VGG16 is derived from its name by the fact that it includes a total number of 16 layers of variable weights and approximately 138 million features.

This model is not only applicable for diagnosing pneumonia but also able to identify tuberculosis, all types of cancer, HIV, arthritis, typhoid, malaria, covid-19, migraine, and a variety of many other diseases [10][12][13].The complete framework of VGG16 is shown below in Fig. 1.

VGG16: The VGG16 architecture always takes an input of 224*224 pixels [19][20].There is a stack of convolutional layers through which the input images are passed and a filter of size 3*3 is used for the process. Along with this, it also uses a 1*1 convolution filter and the stride of the convolution is always fixed to 1 pixel. With almost every stack of a convolution layer, there is always a pooling layer used. The pooling layer is performed on the architecture of 2*2 pixels, with a stride of 2 [18]. The number of dense layers can be used at the end of the architecture based on the requirement. The ReLu is utilized as an activation function in the VGG16 architecture.

The preprocessing of the images is done to normalize the RGB value of each pixel. In this process, every pixel is subtracted from the average value [21]. In VGG16, each convolution has 64 filters along with a padding of 1 pixel. The output of the framework is always of the same size as applied to the network, i.e. 224*224. In the initial stage of the model, the image is passed through the first stack containing two convolution layers with the size of 3*3 following the ReLu function where both the convolutional layers have a number of 64 filters [19][21]. After convolutional layers, the output is passed through the pooling layer which gives output as half of the input to the convolutional layers i.e. 112*112*64.

Now in the second step, the number of filters becomes 128 having an input of 112*112 [21]. After the second stack, a max-pooling is applied to the data which results, in the output as half of the input to the stack i.e. 56*56 containing 128 filters. In the third stack, the same process is applied which results in the output as 28*28*256. In each step, the size of the data gets reduced by half, whereas the filters' size becomes two times the previous number of filters. The same flow goes on till the final pooling layer and results in the output size of 7*7*512. Lastly, three fully connected layers are used to get the prediction output in the output layer.

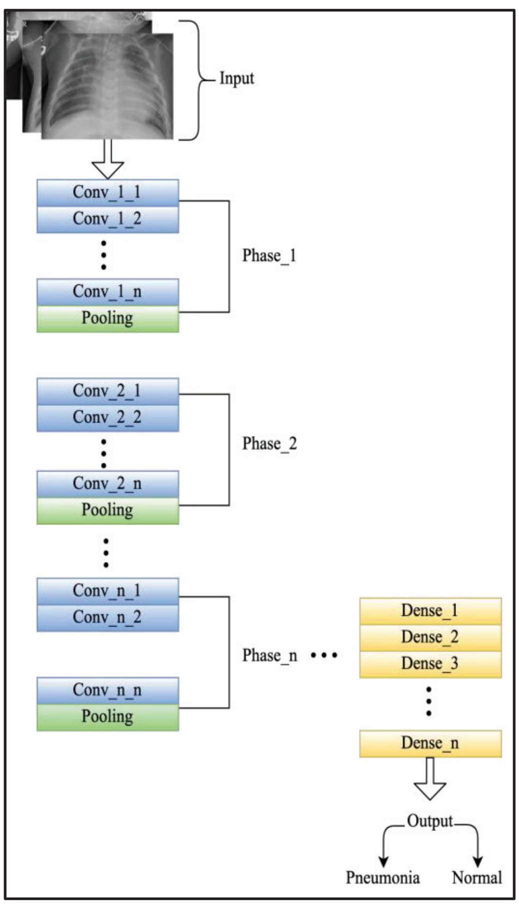

Fig. 1. VGG16 framework

A. Dataset Description

The dataset has been collected from kaggle[22] and is categorized into two folders namely: train, and test; each contains two subdirectories, one of which consists of pneumonia radiographs and the other has normal lung radiographs. A total of 5,856 radiographs are collected from previous pediatric patients aged under 1 to 5 years [23]. The actual data categories were revised and combined, and the complete data set was then restructured for train and test purposes. As a result, the metric of data is applied to the entire dataset.

For testing the model, 4,685 radiographs were assigned to the training phase and 1171 radiographs to the testing process. The dataset is publicly available worldwide in JPEG format.

B. Methodology

The VGG16 transfer learning model is divided into various components such as image augmentation, feature extraction [24], an oversampling technique for imbalanced data, training, testing and pneumonia prediction. In the initial stage, the number of objects for each class in the training dataset has been visualized. In which the results have shown that the training dataset is imbalanced having a huge amount of data for pneumonia and very less data for normal lungs. In the second phase, the test data has been also visualized and identified that it is balanced for both pneumonia and normal lung images. In the third step, the training dataset has been balanced using a data augmentation technique using a number of images of pneumonia and normal lungs from the testing dataset. Thereafter, the shape of the images has been identified using the shape() function, which has shown that the random images picked from the dataset have different shapes, resolutions and channels [25].

Data Preprocessing: In this phase, the data normalization has been done for keeping all the images in the same range through which it is easy for the VGG16 transfer learning model to understand the images [25]. Along with this, the balancing of the training dataset is also performed for getting accurate results and avoiding the problem of overfitting.

Data Augmentation: The technique of developing new training datasets from existing datasets is known as image augmentation [1][6]. In this process, the alteration of the original image takes place by making some changes in the existing sample to create a new one [6]. For example, the new image that is somewhere brighter than the original one can be cut into a subsection; also the filters can be applied to the original images including zoom in, zoom out, shifts and so on. It may generate an almost limitless number of additional training images by performing various image augmentation techniques to the original images present in the training datasets. The feature of applying the image augmentation technique is to avoid overfitting issues, along with this, it can also increase the size of training data. In this step, the images have been resized to 224*224 size for feeding it to the VGG16 model and then converting to PyTorch tensor. Furthermore, the same set of transformations is applied to the test dataset and the rotation on the images has been performed i.e. (-10, 10) degree rotation. Random horizontal, vertical shift along with horizontal flip is applied with a chance of 50%. The image normalization is also applied to convert from ToTensor() function [0-255] dimensions to torch.FloatSensor() function ranges [0.0 to 1.0].

Oversampling technique implementation: As the training dataset has been found imbalanced, hence the oversampling technique is implemented to avoid the issue of overfitting and underfitting. Afterwards, the different layers to be included in the model have been selected. Furthermore, a custom dataset is created which is used to store the data frame and transformation for the images.

Model training and testing: In this phase of the architecture, the complete model is trained on the training dataset. Depending on the value of validation loss, it helps in selecting the best model state. This step results in showing and displaying a loss and metric values at the end of every epoch as shown in Fig. 2.

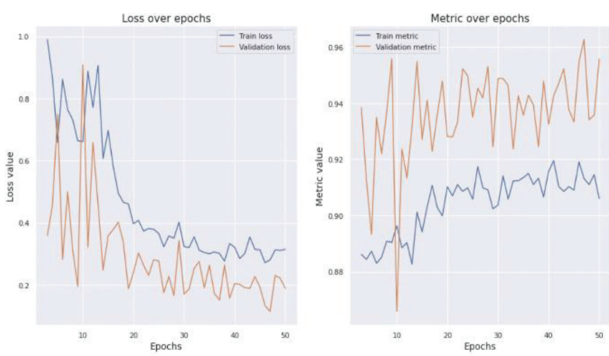

Fig. 2 Loss over epochs Vs Metric over epoch

The total number of epochs has been taken as 50, whereas, the results have been calculated best on the 47th epoch.

Furthermore, the results have been calculated using accuracy, precision, recall, and F1-score performance metrics. In the testing phase, the model has been tested using the test dataset, and the results have been found as accuracy, precision, recall and F1-score.

IV. Results and Discussion

This section compares the performance of the VGG16 transfer learning model with other existing models mentioned in the state-of-the-art section in terms of their accuracy, precision, recall and F1-score parameters.Accuracy is a parameter for predicting the performance of the ML and DL models. It is the percentage of predictions that the model has predicted correctly.

Fig. 3 shows the performance comparison, in which the accuracy has been taken for identifying the best pneumonia prediction model.

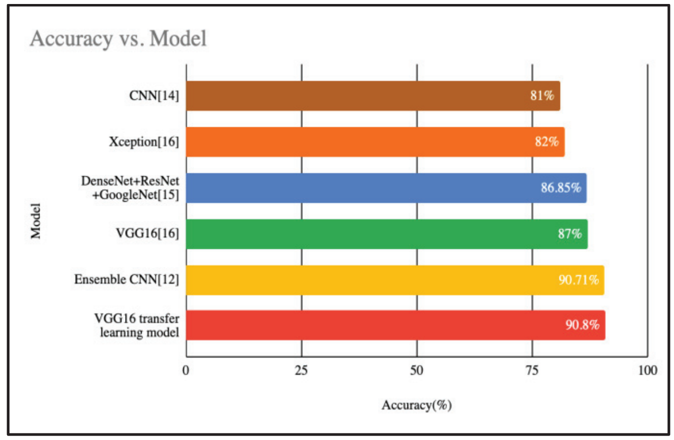

Fig. 3 Accuracy comparison of various models

Fig. 4 shows the comparison of precision among the VGG16 transfer learning model with the pre-existing models. The comparison concludes that the ensemble CNN [12] and the VGG16 transfer learning model as the best models for pneumonia prediction.

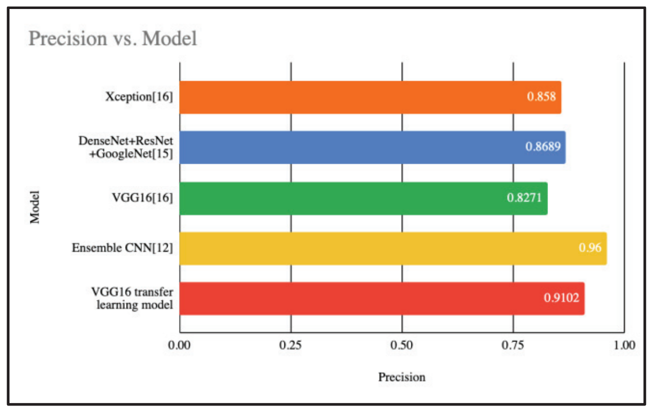

Fig. 4 Precision comparison of various models

Fig. 5 shows the recall of precision among the VGG16 transfer learning model with the pre-existing models. This comparison identifies the VGG16 transfer learning model as the best model for predicting pneumonia with the highest value of recall.

Fig. 6 illustrates the F1-score comparison of the VGG16 transfer learning model with the existing model which identifies that the VGG16 transfer learning model performs best among all due to the highest value of F1-score. Hence, the VGG16 transfer learning model can be validated and recognized as the better prediction model in terms of accuracy in comparison to the various models proposed by various authors in the literature survey.

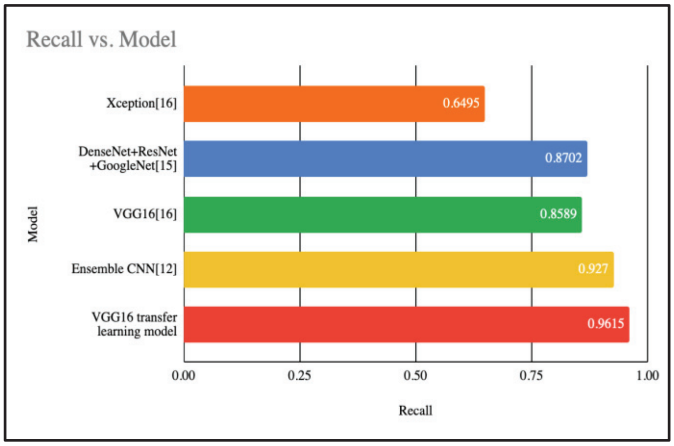

Fig. 5 Recall comparison of various models

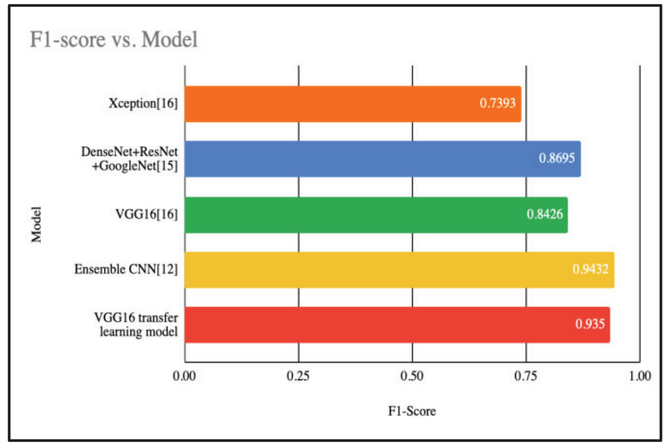

Fig. 6 F1-score comparison of various models

TABLE I. PERFORMANCE COMPARISON OF VARIOUS DL MODELS

Reference	Dataset	Source	Accuracy	Precision	Recall	F1-score
Ensemble CNN[12]	Kermany's and Goldbaum dataset	Kaggle	90.71%	0.96	0.927	0.9432
CNN[14]	Chest X-ray dataset	Advanced Diagnostic and Therapeutic Technology Department of the Grande Ospedale Metropolitano	81%	NA	NA	NA
DenseNet+ResNet+ GoogleNet[15]	RSNA	Kaggle	86.86%	0.8689	0.8702	0.8695
VGG16[16]	Kermany's Dataset	Kaggle	87%	0.8271	0.8589	0.8426
Xception[16]	Kermany's Dataset	Kaggle	82%	0.858	0.6495	0.7393
VGG16 transfer learning model	Chest X-ray images	Kaggle	90.8%	0.9102	0.9615	0.935

Table. I shows the comparison of the VGG16 transfer learning model with the pre-existing DL models in terms of accuracy, precision, recall and F1-score.

V. CONCLUSION AND FUTURE SCOPE

Pneumonia is a respiratory infection which causes alveoli in the lungs to become inflamed. Cough, fever and breathing issues can occur when the air sac fills with fluid. Numerous species, including germs, viruses, bacteria, and fungi, are capable of causing pneumonia. It can range in intensity from non-threatening to fatal. The most vulnerable include young children and newborns, people over 65, and people with bad health conditions or weakened immune systems. There are various ways of diagnosing pneumonia, i.e. using chest x-ray images. But, for this kind of situation, there is a requirement of a specialist. Early identification of pneumonia can help in reducing the mortality rate, which is quite difficult to do so. Hence, in this work, a CNN model with VGG16 is introduced for classifying pneumonia chest x-ray images. The dataset used for the study is divided into two directories of training and testing datasets. Further, the train and test data have two more folders containing the images of pneumonia and normal lungs. The study depicts the results in the form of accuracy, precision, recall, and F1-score as 90.8%, 0.9102, 0.9615, and 0.935, respectively.

In the future, the dataset can be increased which may lead to better performance results. Furthermore, the ensemble of DL models can be used to increase the effectiveness of feature extraction along with better results.

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